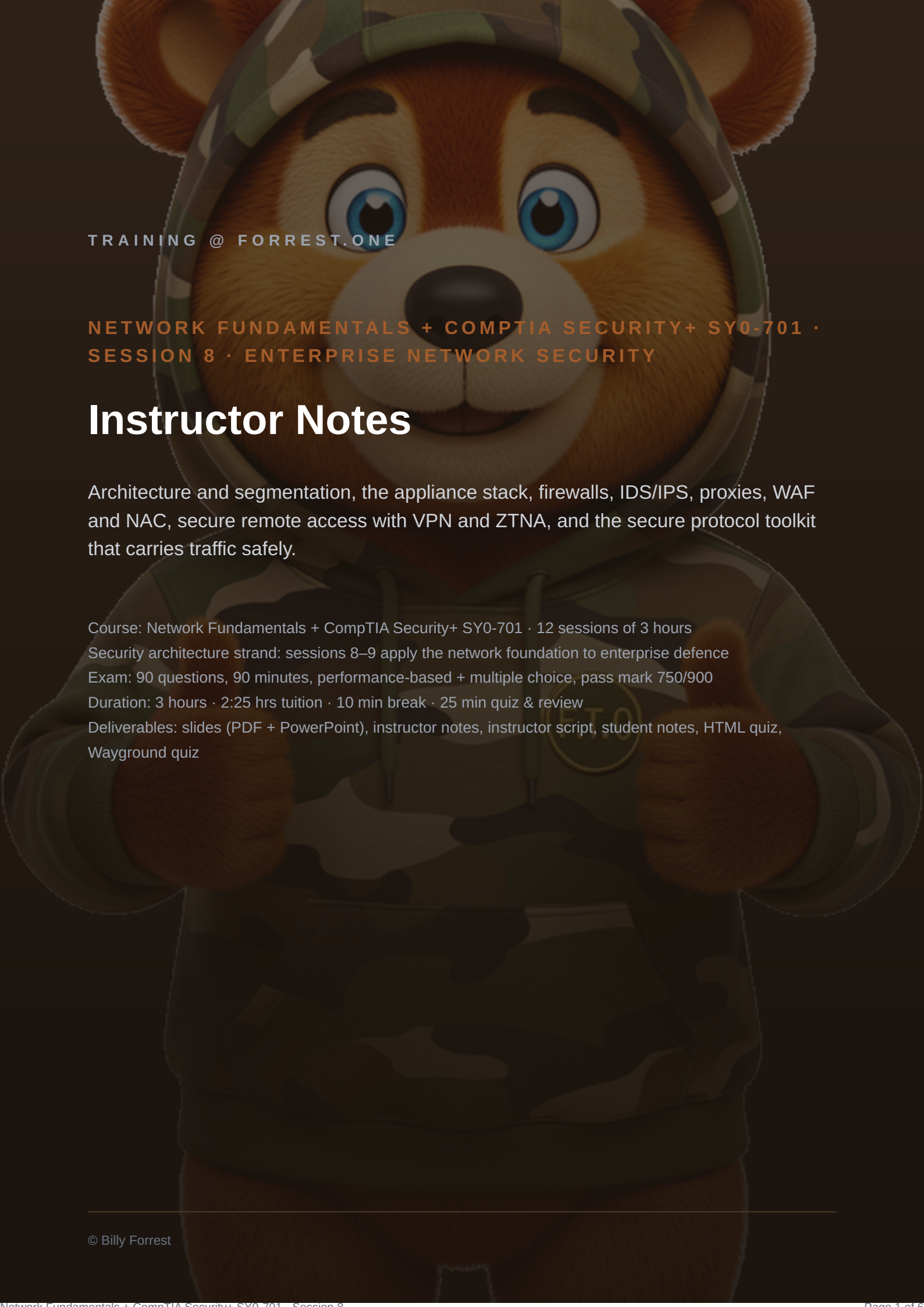




TRAINING @ FORREST.ONE

NETWORK FUNDAMENTALS + COMPTIA SECURITY+ SY0-701 ·
SESSION 8 · ENTERPRISE NETWORK SECURITY

Instructor Notes

Architecture and segmentation, the appliance stack, firewalls, IDS/IPS, proxies, WAF and NAC, secure remote access with VPN and ZTNA, and the secure protocol toolkit that carries traffic safely.

Course: Network Fundamentals + CompTIA Security+ SY0-701 · 12 sessions of 3 hours

Security architecture strand: sessions 8–9 apply the network foundation to enterprise defence

Exam: 90 questions, 90 minutes, performance-based + multiple choice, pass mark 750/900

Duration: 3 hours · 2:25 hrs tuition · 10 min break · 25 min quiz & review

Deliverables: slides (PDF + PowerPoint), instructor notes, instructor script, student notes, HTML quiz, Wayground quiz

Instructor Notes — Enterprise Network Security

A 3-hour delivery guide for Session 8, aligned to the expanded narration: six chapters covering the physical and human layer, segmenting the network, guests and personal devices, hardening and admission control, keys and content control, and trust zones. Keep a placement mindset throughout: for every control, name what it does, when it acts, and where it sits. Anchor everything to defence in depth: physical controls protect the rooms, segmentation contains the blast radius, admission control decides who joins, keys and filters protect the data, and zones describe the whole picture. The activities force the scenario reasoning the exam rewards.

Session at a glance

Time	Block	Slides	Focus / activity
0:00–0:10	Welcome & objectives	1–3	Where this session sits; objectives
0:10–0:32	Physical, deception and audit	4–9	Rooms, surveillance, honeypots, audits, regimes
0:32–0:55	Architecture and segmentation	10–16	Models, segmentation principles, IoT, Purdue model
0:55–1:03	Discussion: IoT trust	17	Why sensors get their own segment
1:03–1:20	Guests and personal devices	18–21	Guest VLANs, captive portal, BYOD
1:20–1:30	Break	22	10-minute break
1:30–1:50	Hardening and admission	23–28	Attack surface, baselines, NAC, 802.1X
1:50–2:05	Keys and content control	29–33	Key lifecycle, key stores, ACLs, filtering, DLP
2:05–2:18	Trust zones	34–35	Zones, screened subnet, depth, zero trust
2:18–2:35	Exam reference and activities	36–40	Appliance and protocol reference; two activities

Time	Block	Slides	Focus / activity
2:35–3:00	Quiz, takeaways and close	41–44	Quiz_Session_08.html; review; preview

Materials and preparation

- Projector or screen with the Session 8 slide deck (02_Slides.pdf or 02_Slides.pptx).
- One computer or tablet per pair for Quiz_Session_08.html (any browser, no internet required).
- Printed or digital copy of the Student Notes for each candidate.
- Whiteboard for the zone diagram and the Purdue layer diagram.
- Optional: a browser's certificate viewer for the TLS and HTTPS reference slide.
- Optional: printed extracts of a CIS benchmark and a vendor hardening guide, to show a real baseline.

Delivery guidance by slide

Slides 1–3 · Welcome and objectives (10 minutes)

Anchor the session in sessions 1–3 and 4–7: VLANs from session 2 are now containment controls, cryptography from session 6 becomes key management, and SSH from session 3 returns in the protocol toolkit. Set the placement mindset: the exam describes a situation and asks which control belongs there.

Slides 4–9 · Physical, deception and audit (22 minutes)

Start at the door. Rooms, closets and locks (slide 4) carry the argument that physical access defeats every logical control, so ask the class to name every route into their own building before you show the slide. Surveillance and visitor control (slide 5) is about proof: coverage, retention and logs are what you reach for during an investigation. Honeypots and honeynets (slide 6) need the isolation point made twice: a decoy that can reach production is a foothold you gave away. Slide 7 is the legal lens, and it is the one students most often forget: consent, privacy and entrapment. Keep it short and concrete. Audits (slide 8) asks a harder question than a policy review, so stress evidence over assertion. The regimes table (slide 9) is where data locality lands, and it decides which cloud regions are even permissible.

Teaching tip

For every physical control, ask: what does it stop, what does it record, and who reviews the record? A camera nobody watches is a liability, not a control.

Slides 10–16 · Architecture and segmentation (23 minutes)

Architecture first (slide 10): tiered, spine-leaf and SMB models, with redundancy as the companion to reliability, since exam scenarios often test a network where one device fails. Then the narration's segmentation principles: why we divide the network (slide 11), planning from data sensitivity and trust level (slide 12), and the mechanisms in practice (slide 13), where VLANs, the DMZ with its twin firewalls and micro-

segmentation are the containment story. The special environments are the heart of this chapter: IoT and industrial systems (slide 14), the Purdue model (slide 15), and the availability-first argument for operational technology (slide 16). Draw the Purdue layers on the whiteboard and ask where an air gap belongs.

Activity (slide 17, discussion)

Sensors and controllers: weak patching, default credentials, no authentication, and uptime that outranks confidentiality. Compromised, they become pivot points. Segment by trust, not convenience.

Slides 18–21 · Guests and personal devices (17 minutes)

Guests get internet only, with no route inward (slide 18), and the captive portal (slide 19) is the control point that records agreement and gives you a log entry. BYOD (slide 20) is the same problem with a personal twist: you do not own the hardware. Slide 21 is the programme: define compliance, enrol or deny, issue a certificate, separate work data, and be ready to wipe corporate data from a lost device without touching the owner's photographs.

Slides 23–28 · Hardening and admission control (20 minutes)

Hardening is simple in principle and neglected in practice. Reducing the attack surface (slide 23) starts with default credentials, then unused services, ports and protocols. Baselines (slide 24) introduce the documented standard and drift, and slide 25 is the exam trap: build the baseline from CIS benchmarks, NIST publications and vendor guides, not from memory. Then admission: posture assessment (slide 26) covers antivirus currency, patches, settings and credentials, with allow, quarantine and deny outcomes plus a remediation network; 802.1X (slide 27) is the mechanism, with the supplicant, authenticator and RADIUS roles; and slide 28 is the deployment reality, monitor mode first, then segment by segment.

Slides 29–33 · Keys and content control (15 minutes)

Cryptography is only as strong as its keys. The key lifecycle (slide 29) is generation, distribution, storage, rotation, revocation and destruction, and the storage options (slide 30) are examined: hardware security module, key vault, secure enclave and key management system. Rotation (slide 31) limits the blast radius. Then written policy becomes enforcement: access control lists and URL and DNS filtering (slide 32), where rule order and first-match behaviour decide everything, and inspection (slide 33), which separates content filtering, deep packet inspection, data loss prevention and the web proxy. Filtering decides what is allowed; inspection decides what is safe.

Teaching tip

Use the ACL example: a permit everything rule at the top of the list makes every rule below it decoration. Ask students to write the order for a guest VLAN by hand.

Slides 34–35 · Trust zones (13 minutes)

Trust zones (slide 34) put architecture around everything else: untrusted, trusted and semi-trusted zones, and the screened subnet as a buffer holding public facing services. Slide 35 adds zone based policy with deliberate asymmetry (internal users reach the DMZ, the DMZ cannot reach back in), defence in depth, and zero trust, which discards the assumption that anything inside the perimeter is safe.

Slides 36–40 · Exam reference and activities (17 minutes)

These slides consolidate the 8B, 8C and 8D exam-objective strand that the session quiz is tagged against: the appliance stack and detection (slide 36), remote access (slide 38) and the secure protocol toolkit (slide 40). They are deliberately dense reference material, so do not read every cell aloud: walk the columns and let students scan. Run both activities (slides 37 and 39); each takes about eight minutes in pairs and four minutes in debrief, and the debrief question 'what happens when each fails' is how exam scenarios are written.

Activities (slides 37 and 39)

Appliance activity: WAF for web apps; IPS blocks inline; load balancers need redundancy, since a failed one is a single point of failure; proxies mask internal topology. Remote access activity: MFA, firewall allowlists, patching and monitoring; site-to-site VPN for branches; SSL VPN for browser-only contractor access. The non-negotiable control is MFA.

Slides 41–44 · Quiz and close (25 minutes)

Run the quiz, review the low scores, walk the takeaways and assign the checklist. Preview Session 9: endpoint, application, cloud and virtualisation security.

Common misconceptions and corrections

Misconception	Correction
"A honeypot is entrapment by another name"	A decoy observes and records; it must never solicit an attack, and monitoring needs consent
"A honeypot can sit on the production VLAN"	Isolation is mandatory: a decoy that can reach production is a foothold
"Physical security is the facilities team's problem"	Physical access defeats every logical control, so it belongs in the security design
"IoT is just another endpoint"	Cheap, rarely patched, default credentials and no authentication: untrusted by default
"Industrial systems should be hardened like servers"	You often cannot patch or reboot them: contain the segment instead
"Guest Wi-Fi only needs a password"	Guest means its own VLAN, no route inward, a bandwidth cap and a portal that records agreement
"BYOD means no policy"	Compliance definition, enrolment, certificate, data separation and selective wipe
"Hardening is installing antivirus"	Hardening is removing what the device does not need, then auditing against a baseline
"A baseline written in-house is good enough"	Use CIS benchmarks, NIST publications and vendor guides, then verify compliance

Misconception	Correction
"NAC is just guest Wi-Fi onboarding"	NAC enforces endpoint posture across the whole network, with quarantine and remediation
"A key never needs to change"	An unrotated key is a key an attacker keeps forever: rotate, revoke, then destroy
"ACL order does not matter"	Rules are first-match: a permit everything rule at the top makes the rest decoration
"IDS and IPS are interchangeable"	IDS detects and alerts; IPS sits inline and blocks
"Stateful means the firewall encrypts"	Stateful tracks connection state; encryption is a separate control
"The DMZ is where backups live"	The DMZ hosts public-facing services so their compromise stays outside
"Anything inside the perimeter is trusted"	Zero trust verifies identity and context on every request, wherever it comes from

Assessment check

- Session objectives map to topic areas 8A–8H and the security-architecture strand.
- The HTML quiz (20 questions, tagged 8A–8D) covers architecture, appliances, remote access and secure protocols; the Wayground quiz (15 questions) supports the in-class assessment.
- Areas 8E–8H (physical, deception and audit; segmentation and special environments; guests and personal devices; key management and content control) are checked through the discussion, the two activities and the self-check questions in the student notes.
- Watch for scenario items that combine controls: segmentation plus NAC plus filtering plus a screened subnet is a defence-in-depth story.